

Organic Synthesis

Highly Chemo-, Enantio-, and Regioselective Synthesis of α,α -Disubstituted Furanones by Cu-Catalyzed Conjugate AdditionKohei Endo,^[a] Sayuri Yakeishi,^[c] Ryotaro Takayama,^[c] and Takanori Shibata^{*,[c]}

Abstract: A highly chemo-, enantio-, and regioselective synthesis of furanones bearing an α,α -disubstituted quaternary stereogenic center is reported. The Cu-catalyzed enantioselective conjugate addition of organoaluminum reagents to unsaturated ketoesters at room temperature and subsequent lactonization took place. Synthetic transformations of furanones represent facile approaches to various cyclic or acyclic compounds bearing a quaternary stereogenic center.

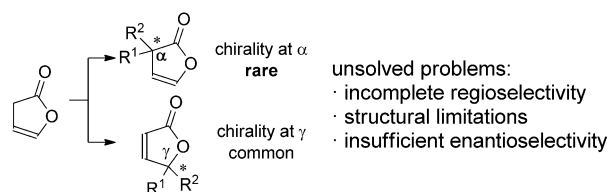
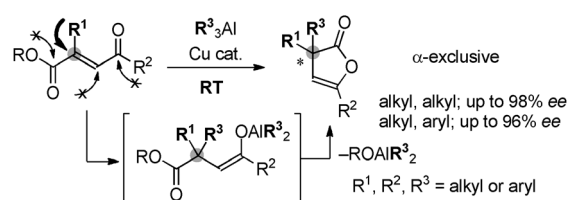


Figure 1. All-carbon chiral quaternary stereogenic centers in butenolides ($R^1, R^2 \neq H$).

Carbonyl compounds bearing an α -chiral carbon atom are ubiquitous building blocks for the synthesis of various biologically active compounds.^[1] There have been many reports on asymmetric alkylation reactions of carbonyl compounds; especially, butenolides, which have been recognized as attractive synthetic targets.^[2] While enantioselective alkylation of the γ -position of butenolides has been extensively studied, there have been few reports of the enantioselective and regioselective alkylation at the α -position. Since α - and γ -alkylation reactions via metal enolate intermediates derived from furanones are typically competitive, regio- and enantioselective alkylation should be difficult to achieve (Figure 1). The use of benzofuranones naturally gives α -alkylation or acylation products; however, the benzofused products are structurally limited.^[3,4] We report here a novel approach to the synthesis of furanones by a Cu-catalyzed chemo-, enantio-, and regioselective conjugate addition reaction (Scheme 1).

Since the chemoselective asymmetric conjugate addition takes place at the β -position of the ketone moiety, furanones



Scheme 1. Present work.

would be obtained via an enolate intermediate. However, the chemoselectivity for the creation of a chiral quaternary carbon rather than a tertiary carbon atom is difficult to control.^[5,6] Furthermore, 1,2-addition reactions of hard Lewis acidic organoaluminum reagents to ester and ketone moieties might take place. The precedent chemo- and enantioselective conjugate addition showed the use of cyclic ketoesters and organozinc reagents at low temperature (Scheme 2a).^[7a] During our original studies, Hoveyda et al. reported the reaction using acyclic enones including two examples of ketoesters and organoaluminum reagents to give simple acyclic ketoesters in moderate yields with good enantioselectivities; unexpectedly, furanones were not described (Scheme 2b).^[7b,c] Fillion et al. reported the synthesis of lactones by the enantioselective conjugate addition of organozinc reagents to Meldrum's acid derivatives, but a few steps for the transformations were required and furanones could not be synthesized (Scheme 2c).^[7d-f] We hypothesized the tandem reaction through the enantioselective addition to acyclic ketoesters for the synthesis of furanones. Typically, the synthesis of furanones bearing an α,α -alkyl,alkyl quaternary stereogenic center has been developed via enolate analogues. Enantioselective decarboxylative allylation via a carbonate derivative is a useful approach, but the regio- and enantioselectivity needs to be further improved (Scheme 2d).^[8,9] A chiral organocatalyst for asymmetric migration of carbonate derivatives gave chiral β -dicarbonyl compounds and δ -dicarbonyl compounds as mixtures; the severe

[a] Prof. Dr. K. Endo
Department of Chemistry, Graduate School of Natural Science
and Technology, Kanazawa University
Kakuma, Kanazawa, Ishikawa, 920-1192 (Japan)
E-mail: kendo@se.kanazawa-u.ac.jp

[b] Prof. Dr. K. Endo
PRESTO (Japan) Science and Technology Agency (JST)
4-1-8 Honcho, Kawaguchi, Saitama, 332-0012 (Japan)

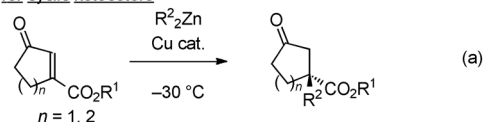
[c] S. Yakeishi,⁺ R. Takayama,⁺ Prof. Dr. T. Shibata
Department of Chemistry and Biochemistry
Graduate School of Science and Technology
Waseda University, Ohkubo, Shinjuku
Tokyo, 169-8555 (Japan)
E-mail: tshibata@waseda.ac.jp

[*] S.Y. and R.T. contributed equally to this work.

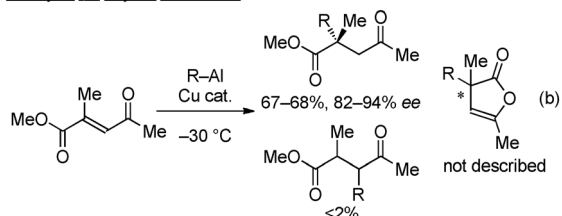
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Addition to unsaturated ketoesters

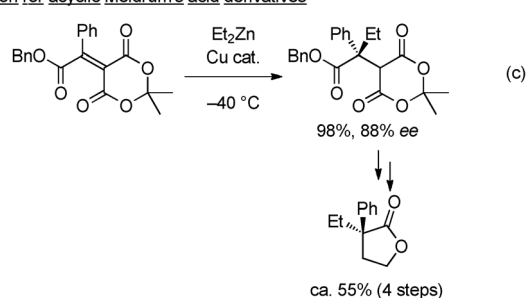
Hoveyda for cyclic ketoesters^[7a]



Hoveyda for acyclic ketoesters^[7b,c]

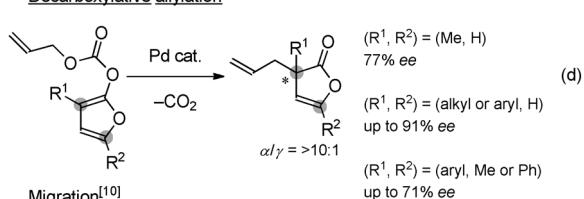


Fillion for acyclic Meldrum's acid derivatives^[7d–f]

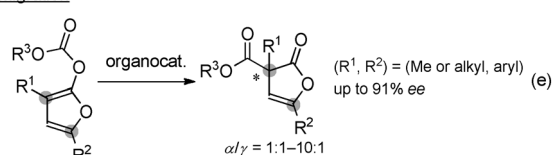


Previous approaches to furanones

Decarboxylative allylation^[8,9]



Migration^[10]



Scheme 2. Addition to unsaturated ketoesters and previous approaches to furanones.

structural limitations and insufficient enantioselectivity are still problematic (Scheme 2e).^[10]

We envisaged the conjugate addition/intramolecular lactonization as a practical and regioselective synthesis of furanones bearing a quaternary stereogenic center. To achieve our goal, acyclic α,β -unsaturated ketoesters were designed as acceptors for organometallic reagents. Our laboratory recently developed an original multinuclear system for Cu- or Pd-catalyzed asymmetric addition reactions of organometallic reagents.^[11] The original Cu catalyst achieved the highly enantioselective conjugate addition of organoaluminum reagents to β,β -disubstituted enones for the creation of a quaternary stereogenic center.^[11d] We envisaged that the present approach could be applied to unsaturated ketoesters to give chiral furanones. The

difficulty in the present reaction is the chemoselective conjugate addition to create a quaternary stereogenic center. Generally, while conjugate addition to create a tertiary stereocenter and use of cyclic enones has been comprehensively studied, conjugate alkylation to acyclic enones to create an all-carbon quaternary stereogenic center is still a challenging topic.

We examined the use of our original Cu complexes in the chemoselective conjugate addition of organoaluminum reagents to unsaturated ketoesters.^[12] The present reaction proceeds by the asymmetric conjugate addition of organoaluminum reagents, followed by the intramolecular trapping of an aluminum enolate with an ester moiety to give the desired furanones exclusively without the generation of a regioisomer. Our original ligands, **BmP**, **BP**, and **SP** (Figure 2), were examined for the present reaction (Table 1). Optimization of the reaction conditions revealed that the use of benzyl ester **1a** in the presence of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (5 mol%) and **SP** (5 mol%) in THF at 0 °C to RT gave the desired product **2a** in high yield with excellent enantioselectivity, although Hoveyda et al. reported the reaction using acyclic unsaturated ketoesters gave acyclic

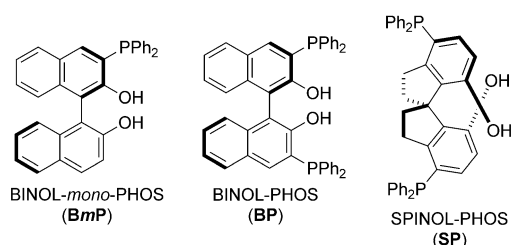


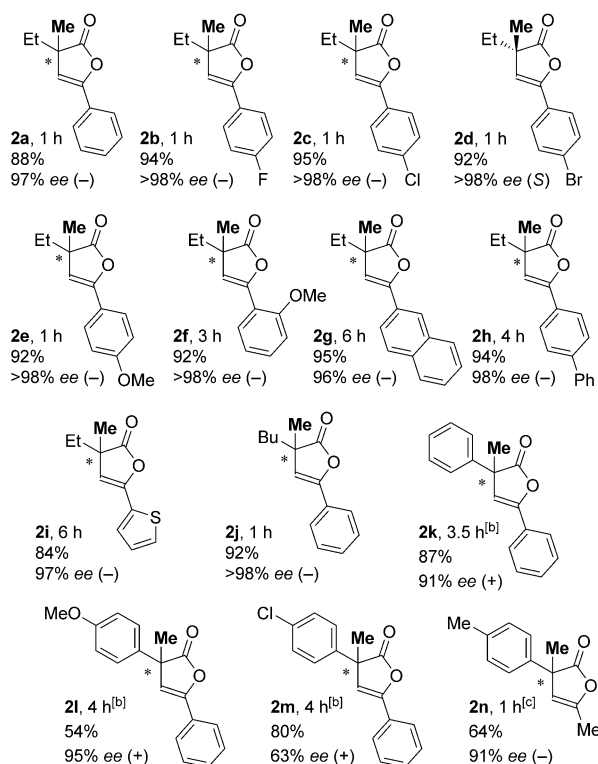
Figure 2. Representative ligands.

Table 1. Conditions for Me_3Al .

Entry	R	Ligand (X)	Cu salt	t [h]	Yield, ee [%]
1	Me (1a-Me)	BmP (10)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	2	80, 52 (+)
2	Me (1a-Me)	BP (5)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	4	58, 36 (+)
3	Me (1a-Me)	SP (5)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	1	82, 83 (–)
4	Et (1a-Et)	BmP (10)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	1	56, 69 (+)
5	Et (1a-Et)	BP (5)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	1	93, 70 (+)
6	Et (1a-Et)	SP (5)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	1	74, 90 (–)
7	Bn (1a)	BmP (10)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	2	90, 70 (+)
8	Bn (1a)	BP (5)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	2	81, 36 (+)
9	Bn (1a)	SP (5)	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	1	80, 96 (–)
10	Bn (1a)	SP (5)	$\text{Cu}(\text{OTf})_2$	2	69, 97 (–)
11	Bn (1a)	SP (5)	$\text{Cu}(\text{acac})_2$ ^[b]	2	74, 98 (–)
12	Bn (1a)	SP (5)	$\text{Cu}(\text{OAc})_2$	2	81, 98 (–)
13	Bn (1a)	SP (5)	$\text{CuBr} \cdot \text{SMe}_2$	16	77, 98 (–)
14 ^[a]	Bn (1a)	SP (5)	$\text{Cu}(\text{OAc})_2$	1	88, 97 (–)

[a] Me_3Al (1.5 equiv) was used. [b] acac = acetylacetonate.

γ -ketoesters (Table 1, entries 1–9). BINOL derivatives **BmP** and **BP** or the SPINOL derivative **SP** gave opposite major enantiomers, respectively.^[11c,d] The screening of Cu salt showed that the use of Cu(OAc)₂ gave the product in better yield with excellent *ee* values (entries 10–14). The optimized reaction conditions were used for a variety of benzyl ketoesters (Scheme 3).^[13] Benzoyl moieties bearing an electron-withdrawing group, such as F, Cl, or Br, or electron-donating group were compatible; the products **2a–f** were obtained in high to excel-



Scheme 3. List of products for Me₃Al. [a] For conditions, see Table 1, entry 14. [b] **BmP** (10 mol %) and Me₃Al (2 equiv) were used. [c] The corresponding ethyl ester, Cu(OAc)₂ (5 mol %), **BmP** (10 mol %), Me₃Al (6 equiv), and ether as a solvent were used.

lent yields with high *ee* values. The absolute configuration of **2d** (S) was deduced by X-ray single-crystallographic analysis.^[14] Electron-rich substituents, such as 2-naphthyl, 4-biphenyl, and thiophen-2-yl, were also compatible and the products **2g–i** were obtained. The reaction using a ketoester bearing a longer alkyl group gave the product **2j** in 92% yield with more than 98% *ee*. The aryl-substituted ketoesters gave the products **2k–n** with high *ee* values when **BmP** was used as a ligand; the present reaction provides an alternative approach to the α -arylation of carbonyl compounds.

Conjugate addition of ethyl- and phenylaluminums was examined as representative for alkyl- and aryl-addition reactions (Table 2). The addition of Et₃Al required Me₂AlCl and AgOAc as additives to give **2o** in high yield with high *ee* (Table 2, entries 1–7).^[15] The typical transition-state model according to the literature suggested that Lewis acid additives could change

Table 2. Conditions for Et₃Al.

Entry	Cu salt	<i>t</i> [h]	Yield, <i>ee</i> [%] ^[a]
1	Cu(OAc) ₂	6	60, 80
2	CuCl ₂ ·2H ₂ O	24	71, 77
3	Cu(OTf) ₂	21	62, 74
4	Cu(acac) ₂	21	66, 82
5 ^[b]	Cu(OAc) ₂	6	52, 74
6 ^[c]	Cu(OAc) ₂	6	79, 84
7 ^[c]	Cu(acac) ₂	4	74, 85 ^[d]

[a] NMR spectroscopic yield. [b] Me₂AlCl (20 mol %) was added. [c] Me₂AlCl (20 mol %) and AgOAc (20 mol %) were added. [d] Isolated yield.

the catalytic performance (Figure 3).^[16] Furthermore, conjugate addition of phenylaluminum to obtain α -arylfuranones was examined; the intermolecular α -arylation of carbonyl compounds is important, but has not yet been well established for further transformations.^[17] The reaction using Ph₃Al prepared in situ from PhLi or PhMgBr and AlCl₃ did not give the product (Table 3, entry 1). In contrast, the reaction using PhEt₂Al or PhMe₂Al gave the desired product **2o** (entries 2 and 3). Optimization of the reaction conditions showed that the use of PhMe₂Al and Cu(OTf)₂ as a Cu salt could give the product **2o**

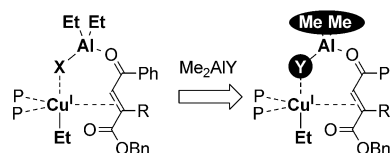


Figure 3. Proposed effect of additives.^[15]

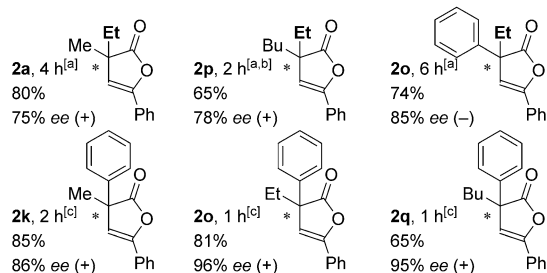
Table 3. Conditions for PhR₂Al.

Entry	R	Cu salt	<i>t</i> [h]	Yield, <i>ee</i> [%] ^[a]
1	Ph	Cu(OAc) ₂	24	no reaction
2	Et	Cu(OAc) ₂	3	49, 72
3	Me	Cu(OAc) ₂	6	67, 85
4	Me	Cu(OTf) ₂	6	42, 95
5	Me	CuCl ₂ ·2H ₂ O	6	38, 83
6	Me	Cu(acac) ₂	2	55, 84
7 ^[b]	Me	Cu(OTf) ₂	1	82, 96

[a] NMR spectroscopic yield. [b] Cu(OTf)₂ (10 mol %) and SP (10 mol %) were used.

in high yield with excellent enantioselectivity (entry 7); the present reaction generated the Et or Me adducts as byproducts, respectively (entries 2–7).^[15] Ketoesters bearing a methyl, butyl, or phenyl group participated in the addition of Et₃Al to give **2a**, **2o**, and **2p** with high ee values (Scheme 4). The addition of PhMe₂Al to ketoesters bearing a methyl, ethyl, or butyl group gave the products **2k**, **2o**, and **2q** with high to excellent ee values, respectively.

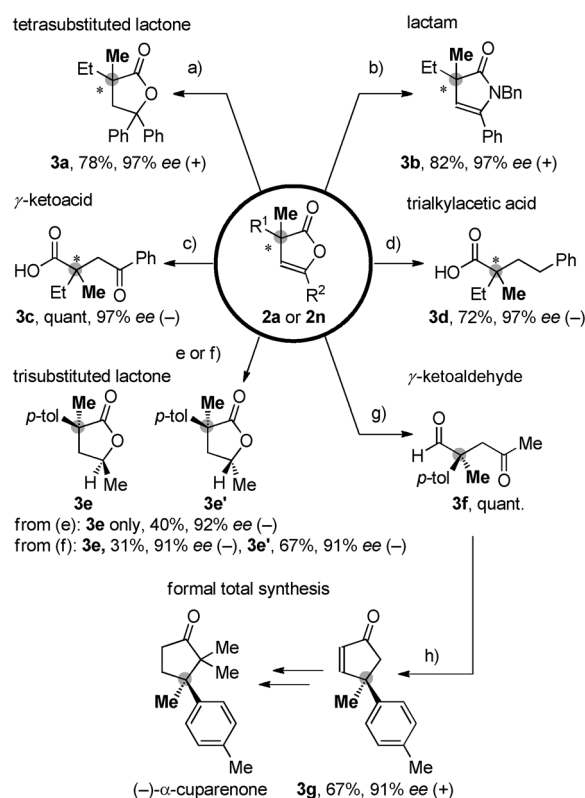
Several transformations of the products were preliminary examined (Scheme 5). A Friedel–Crafts-type reaction of **2a** unexpectedly gave the sterically congested product 4,4-disubstituted



Scheme 4. List of products for Et₃Al and PhMe₂Al. [a] For conditions, see Table 2, entry 7. [b] Cu(OAc)₂ (10 mol %) and **SP** (10 mol %) were used. [c] For conditions, see Table 2, entry 7.

ed γ -lactone **3a** in 78% yield without addition to a carbonyl moiety (Scheme 5a). The amidation of **2a** by using benzylamine in the presence of InCl₃ catalyst under reflux conditions gave the γ -lactam **3b** in 82% yield (Scheme 5b). The treatment of **2a** under basic conditions gave the ring-opening product **3c** in quantitative yield; the present approach would give a wide variety of γ -ketoacids bearing a quaternary stereogenic center (Scheme 5c). We found that the formal reduction of a ketone to a methylene group took place; the Pd-catalyzed hydrogenation and subsequent debenzoylation of **2a** gave the simple carboxylic acid **3d** in 72% yield (Scheme 5d). The hydrogenation of **2n** in the presence of Pd(OAc)₂ gave the corresponding lactone **3e** as a single diastereomer (Scheme 5e). In addition, hydrogenation in the presence of Pd/C gave the desired product **3e** and **3e'** quantitatively as a diastereomeric mixture (Scheme 5f). The reduction of **2n** by using DIBAL-H and the intramolecular cyclization gave cyclopentenone **3g**, which is the synthetic intermediate for cuparenone, as a representative small natural product (Scheme 5g,h).^[18,19] Many natural products bear a methylated quaternary stereogenic center; thus, the present facile synthesis of furanones should be noteworthy as a “synthetic platform” in organic chemistry.

In conclusion, we have developed the first chemo-, regio-, and enantioselective synthesis of furanones bearing an α -chiral quaternary stereogenic center via the asymmetric Cu-catalyzed conjugate addition of organoaluminum reagents to unsaturated ketoesters under ambient conditions. Me-, Et-, and Ph-addition were described along with a demonstration of Lewis acid-controlled catalytic performance. Thus, a wide variety of nucleophiles could take part in the present reaction with high



Scheme 5. a) AlCl₃ (3 equiv), benzene, RT, 4 h; b) InCl₃ (20 mol %), benzylamine, reflux, 24 h; c) 6 N aq. NaOH, reflux, 2 h; d) Pd(OAc)₂ (5 mol %), HCO₂K (3 equiv), DMF, 60 °C, 24 h; e) Pd(OAc)₂ (10 mol %), HCO₂K (5 equiv), DMF, 60 °C, 24 h; f) Pd/C (10 mol %), H₂ (1 atm.), EtOH, 60 °C, 12 h; g) DIBAL-H (1.5 equiv), THF, –78 °C, 2 h; h) KOH, THF/EtOH, RT, 1 h.

enantioselectivities. The synthetic transformations of furanones reflect their versatility as synthetic intermediates. The present inexpensive and easily available Cu catalyst is useful for preparing densely functionalized scaffolds bearing a quaternary stereogenic center. We opened up novel synthetic targets, α,α -disubstituted furanones, via various asymmetric conjugate addition reactions to unsaturated ketoesters; further studies using our original catalysts with a Lewis acid controlled system are underway, and the use of unsaturated ketoesters as Michael acceptors for the synthesis of biologically active compounds will also be reported in the future.

Experimental Section

Representative procedure for Me₃Al

Cu(OAc)₂ (2.3 mg, 0.0125 mmol, 5 mol %) and **SP** (7.8 mg, 0.0125 mmol, 5 mol %) were dissolved in THF (2.5 mL) and the mixture was stirred at room temperature for 30 min and then cooled to 0 °C. A solution of Me₃Al in heptane (0.38 mmol, 0.19 mL, 2 M) was added to the mixture. Ketoester **1a** (73.6 mg, 0.25 mmol) was added to the clear colorless solution at once. The reaction was carried out at RT and monitored by TLC analysis. After the completion of the reaction, a minimal amount of sat. aq. NH₄Cl was added at 0 °C. After stirring at 0 °C for 30 min, the mixture was passed through a pad of silica gel with ether (50 mL). Concentration and

purification by silica gel column chromatography gave the desired product **2a** in 44.5 mg, 0.22 mmol, 88% yield.

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Keywords: addition reactions · aluminum · asymmetric synthesis · copper · heterocyclic compounds

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